

ECON 636 600: MACROECONOMIC THEORY I

Midterm (Oct 16)

1. (40 points) Consider a two period growth model. $u(c) = \log(c)$ and $f(k) = k^\alpha$, $\delta = 1$ and $t = 0, 1$. The planner chooses $\{c_0, k_1, c_1\}$ to maximize

$$\max \quad \log(c_0) + \beta \log(c_1)$$

$$s.t. \quad c_0 + k_1 = 2k_0^\alpha$$

$$c_1 = 2k_1^\alpha$$

$$k_0 \text{ is given.}$$

- (a) Write down the Euler equation.

$$u'(c_0) = \beta u'(c_1) \alpha 2k_1^{\alpha-1}$$

- (b) Solve for the optimal c_0, k_1, c_1 .

Euler equation implies

$$\frac{1}{c_0} = \frac{\beta}{2k_1^\alpha} \alpha 2k_1^{\alpha-1}$$

plug it into the budget constraint $c_0 + k_1 = 2k_0^\alpha$, we get

$$c_0 = \frac{1}{1 + \alpha\beta} 2k_0^\alpha$$

$$k_1 = \frac{\alpha\beta}{1 + \alpha\beta} 2k_0^\alpha$$

$$c_1 = 2\left(\frac{\alpha\beta}{1 + \alpha\beta} 2k_0^\alpha\right)^\alpha$$

2. (a) (15 points) Write down the Bellman equation for the value function.

$$V(s) = \max_{0 \leq s' \leq s} \log(s - s') + \beta V(s')$$

- (b) (25 points) Use the guess that the value function has a form $V(s) = a + b \log(s)$ and determine parameters (a, b) .

$V(s) = \max_{0 \leq s' \leq s} \log(s - s') + \beta(a + b \log(s'))$. The optimal $s' = \frac{\beta b}{1 + \beta b} s$, hence

$$\begin{aligned} V(s) &= \log(s - s') + \beta(a + b \log(s')) = \log\left(\frac{1}{1 + \beta b} s\right) + \beta a + \beta b \log\left(\frac{\beta b}{1 + \beta b} s\right) \\ &= (1 + \beta b) \log(s) + \log\left(\frac{1}{1 + \beta b}\right) + \beta a + \beta b \log\left(\frac{\beta b}{1 + \beta b}\right) \end{aligned}$$

Hence

$$\begin{aligned} b &= \frac{1}{1 - \beta} \\ a &= \frac{\log\left(\frac{1}{1 + \beta b}\right) + \beta b \log\left(\frac{\beta b}{1 + \beta b}\right)}{1 - \beta} = \frac{\log(1 - \beta)}{1 - \beta} + \frac{\beta}{(1 - \beta)^2} \log(\beta) \end{aligned}$$

3. (20 points) In our growth model, the contraction mapping operator is defined as

$$(Tv)(k) = \max_{k' \in [0, F(k, 1) + (1 - \delta)k]} u(F(k, 1) + (1 - \delta)k - k') + \beta v(k').$$

Prove that, if $v(k)$ is weakly increasing in k , then $Tv(k)$ is strictly increasing in k . (You may make reasonable assumptions on production function and utility function when necessary)

Let $k^1 < k^2$ be two initial conditions.

$$\begin{aligned} (Tv)(k^1) &= \max_{k' \in [0, F(k^1, 1) + (1 - \delta)k^1]} u(F(k^1, 1) + (1 - \delta)k^1 - k') + \beta v(k') \\ (Tv)(k^2) &= \max_{k' \in [0, F(k^2, 1) + (1 - \delta)k^2]} u(F(k^2, 1) + (1 - \delta)k^2 - k') + \beta v(k') \end{aligned}$$

Let k^{1*} be the optimal solution in the first problem, then

$$\begin{aligned}
(Tv)(k^1) &= \max_{k' \in [0, F(k^1, 1) + (1-\delta)k^1]} u(F(k^1, 1) + (1-\delta)k^1 - k') + \beta v(k') \\
&= u(F(k^1, 1) + (1-\delta)k^1 - k^{1*}) + \beta v(k^{1*}) \\
&< u(F(k^2, 1) + (1-\delta)k^2 - k^{1*}) + \beta v(k^{1*}) \\
&\leq \max_{k' \in [0, F(k^2, 1) + (1-\delta)k^2]} u(F(k^2, 1) + (1-\delta)k^2 - k') + \beta v(k') = (Tv)(k^2),
\end{aligned}$$

where the first inequality follows the strict monotonicity of production function and utility function, and the second inequality is because $k^{1*} \in [0, F(k^2, 1) + (1-\delta)k^2]$ is a feasible choice in the problem $\max_{k' \in [0, F(k^2, 1) + (1-\delta)k^2]} u(F(k^2, 1) + (1-\delta)k^2 - k') + \beta v(k')$.